Stochastic Calculus and Option Pricing
Lecture 14 Exercises

14 Week 14 — Pricing Toolbox: A Guided Refresher

Focus. A final theoretical variation on the product rule for stochastic processes.

Exercise 14.1 (Quadratic covariation controls terminal covariance)**.** Let X and Y be bounded continuous martingales on $[0, T]$, with deterministic initial values.

(a) Apply the product rule to $X_t Y_t$, keeping the quadratic covariation term explicit.

(b) If $\mathbb{E}[\langle X, Y \rangle_T] = 0$, take expectations and show that $\mathrm{Cov}(X_T, Y_T) = 0$.

(c) Give a simple example of dependent, uncorrelated random variables to explain why zero terminal covariance does not imply independence.